

Low-Rank Decay for Grokking in Scale-Invariant Transformers: A Spectral-Geometric View

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Preliminary Draft

Abstract

Modern Transformer architectures frequently employ normalization mechanisms such as RMSNorm and Query-Key Normalization, making parts of the model approximately scale-invariant with respect to weight magnitudes. In this regime, standard Frobenius-norm weight decay can shrink the radial scale of a parameter matrix without directly simplifying the function represented by the normalized layer. We study grokking in small algorithmic tasks through this lens and propose *Low-Rank Decay* (LRD), a nuclear-norm-like spectral regularizer that penalizes singular values through an efficient Newton–Schulz approximation of the polar factor. On modular addition, preliminary experiments suggest that LRD induces a sharp effective-rank collapse in Query/Key matrices and expands the data-scarce regime in which delayed generalization occurs. We further give a spectral-geometric interpretation: nuclear-norm regularization first drives weights toward low-rank strata, while the enlarged subdifferential near low-rank points leaves task gradients to select the remaining functional subspace. This draft reports early evidence and outlines future experiments needed to establish robustness, causality, and connections to geometry-preserving model merging.

Keywords. Grokking; scale invariance; spectral regularization; nuclear norm; Newton–Schulz iteration; rank collapse; training dynamics.

1 Introduction

Grokking refers to a delayed generalization phenomenon in which a model first memorizes the training set and only much later transitions to near-perfect test accuracy on algorithmic tasks [5]. This behavior suggests that training dynamics are not merely fitting labels, but may involve a transition from a high-capacity memorization solution to a lower-complexity algorithmic circuit.

At the same time, modern neural architectures increasingly rely on normalization layers. In Transformers, RMSNorm [6] and Query-Key Normalization (QK-Norm) normalize activations or attention queries/keys, making the resulting computation insensitive to scalar rescaling of certain weight matrices. This raises a basic question: *if a layer is scale-invariant, what does standard weight decay actually regularize?*

The usual Frobenius penalty $\|W\|_F^2$ controls the magnitude of a weight matrix. However, if the function is invariant under $W \mapsto \alpha W$ for $\alpha > 0$, shrinking $\|W\|_F$ does not directly simplify the function. We therefore explore the hypothesis that, for scale-invariant components, an appropriate inductive bias should act on the *spectral shape* of the matrix, rather than only its scalar norm.

This draft investigates a simple method, *Low-Rank Decay* (LRD), which applies a nuclear-norm-like decay to selected weight matrices. The nuclear norm $\|W\|_* = \sum_i \sigma_i(W)$ acts as a convex

surrogate for rank and encourages spectral sparsity. To avoid per-step singular value decomposition (SVD), we approximate the polar factor $UV^\top$ of $W = U\Sigma V^\top$ using Newton–Schulz iteration and apply a decoupled decay step.

Our preliminary experiments on modular addition suggest that LRD induces rapid effective-rank collapse in attention Query/Key matrices and improves grokking under low data fractions. The main contributions of this draft are:

1. We formulate the mismatch between Frobenius-norm weight decay and scale-invariant Transformer components.
2. We introduce a decoupled Low-Rank Decay update based on a Newton–Schulz approximation of the nuclear-norm subgradient.
3. We present preliminary evidence that LRD induces rank collapse before generalization in modular arithmetic grokking.
4. We propose a spectral-geometric interpretation of LRD near low-rank strata, including the “needle-to-fan” expansion of the nuclear-norm subdifferential.

2 Background and Related Work

Grokking. Grokking was introduced by Power et al. [5] on algorithmic datasets such as modular arithmetic. Later work connected grokking to mechanistic interpretability, showing that generalization can correspond to the formation of structured circuits, for example Fourier or trigonometric circuits in modular addition [4]. Our work focuses on an optimization-side intervention that may accelerate or stabilize the transition toward such low-complexity solutions.

Scale-invariant optimization. A parameter block is scale-invariant if multiplying it by a positive scalar does not change the represented function. Prior work on normalization and scale-invariant parameters has shown that weight decay can alter effective learning rates by shrinking the parameter norm, even when it does not directly impose a functional complexity penalty [1, 3]. We use this as motivation to consider spectral regularization instead of purely radial norm regularization.

Spectral regularization and low-rank bias. The singular spectrum of a weight matrix is a natural complexity measure. Nuclear-norm penalties encourage low-rank solutions and are widely used as convex surrogates for rank. In deep learning, low-rank adaptation methods such as LoRA impose a hard architectural rank constraint [2]. In contrast, LRD is a soft regularizer: it encourages rank reduction but allows task gradients to maintain or regrow singular directions when needed.

3 Scale Invariance and the Limitation of Weight Decay

Definition 1 (Scale invariance). *A function $f(x; W)$ is scale-invariant with respect to W if*

$$f(x; \alpha W) = f(x; W), \quad \forall \alpha > 0. \quad (1)$$

For an attention head with QK-Norm, attention logits take the schematic form

$$A(X) = \text{softmax} \left(\frac{\text{Norm}(XW_Q) \text{Norm}(XW_K)^\top}{\sqrt{d_h}} \right), \quad (2)$$

where $\text{Norm}(z)$ removes the scale of z . Hence scaling W_Q or W_K by a positive scalar does not change the normalized queries or keys.

Let $W = \rho\Theta$, where $\rho = \|W\|_F$ and $\|\Theta\|_F = 1$. Suppose the task loss $\mathcal{L}_{\text{task}}(W)$ depends only on Θ . Consider the continuous-time objective

$$J(W) = \mathcal{L}_{\text{task}}(W) + \frac{\lambda}{2} \|W\|_F^2. \quad (3)$$

The gradient flow is

$$\dot{W} = -\nabla_W \mathcal{L}_{\text{task}}(W) - \lambda W. \quad (4)$$

Since $\mathcal{L}_{\text{task}}$ is homogeneous of degree zero with respect to W , Euler’s homogeneous function theorem gives

$$\langle \nabla_W \mathcal{L}_{\text{task}}(W), W \rangle = 0. \quad (5)$$

Thus the task gradient is tangent to the sphere of constant Frobenius norm, while the weight-decay term is radial.

Proposition 1 (Radial decoupling of Frobenius weight decay). *Under the idealized assumptions above, Frobenius-norm weight decay affects only the radial scale ρ and has no direct projection onto the angular dynamics of Θ .*

Sketch. The radial dynamics satisfy

$$\dot{\rho} = \langle \dot{W}, \Theta \rangle = -\langle \nabla_W \mathcal{L}_{\text{task}}, \Theta \rangle - \lambda \langle W, \Theta \rangle = -\lambda \rho. \quad (6)$$

For the angular variable, the projection of $-\lambda W = -\lambda \rho \Theta$ onto the tangent space at Θ vanishes. Hence the decay term changes ρ but not the angular direction directly. $\square$

Remark 1. *This is an idealized continuous-time statement. In practical training with AdamW, finite steps, non-normalized parameters, and approximate scale invariance, weight decay can still influence optimization indirectly. The claim is not that weight decay is useless, but that it is not a direct functional complexity prior for perfectly scale-invariant parameter blocks.*

4 Low-Rank Decay

4.1 Nuclear-norm objective

We replace the radial Frobenius penalty with a spectral penalty:

$$J_{\text{LRD}}(W) = \mathcal{L}_{\text{task}}(W) + \lambda \|W\|_*, \quad \|W\|_* = \sum_i \sigma_i(W). \quad (7)$$

If $W = U\Sigma V^\top$ is full-rank and has nonzero singular values, one subgradient of $\|W\|_*$ is

$$\nabla_W \|W\|_* = UV^\top = \text{polar}(W). \quad (8)$$

This polar factor preserves the singular-vector geometry while discarding singular magnitudes. A decay step along $UV^\top$ reduces singular values in an ℓ_1 -like manner, encouraging small singular values to vanish.

Algorithm 1 Decoupled Low-Rank Decay via Newton–Schulz

Require: Weight matrix W , learning rate η , decay coefficient λ , iterations K , small $\epsilon > 0$

```
1:  $X \leftarrow W/(\|W\|_F + \epsilon)$ 
2: for  $k = 1$  to  $K$  do
3:    $A \leftarrow X^\top X$ 
4:    $X \leftarrow \frac{1}{2}X(3I - A)$ 
5: end for
6:  $G_{\text{reg}} \leftarrow \lambda X$ 
7:  $W \leftarrow W - \eta G_{\text{reg}}$ 
```

4.2 Newton–Schulz approximation

Exact SVD is expensive inside every optimizer step. We therefore approximate the polar factor using Newton–Schulz iteration:

$$X_0 = \frac{W}{\|W\|_F + \epsilon}, \quad X_{k+1} = \frac{1}{2}X_k(3I - X_k^\top X_k). \quad (9)$$

For rectangular matrices, one can use the corresponding left- or right-oriented variant depending on matrix shape. In practice, a small fixed number of iterations is used.

4.3 Subdifferential geometry near low-rank strata

The nuclear norm becomes non-smooth at rank-deficient matrices. Let $W = U_r \Sigma_r V_r^\top$ be a compact SVD with rank r . The subdifferential is

$$\partial \|W\|_* = \left\{ U_r V_r^\top + Z : U_r^\top Z = 0, Z V_r = 0, \|Z\|_2 \leq 1 \right\}. \quad (10)$$

When W is square and full-rank, the orthogonal complements are trivial and the subgradient is unique. As rank decreases, the zero-singular-value subspaces enlarge, and the admissible Z directions expand. Geometrically, the subgradient changes from a near-unique “needle” direction to a fan-like set over the null singular subspaces.

This suggests the following interpretation of LRD. Early in training, when weights are effectively full-rank, LRD provides a clear polar-factor shrinkage direction. Later, near low-rank strata, the regularizer becomes less prescriptive: it has already removed many weak spectral directions, while task gradients can decide which remaining singular subspaces should be preserved or regrown. This gives LRD a soft-selection behavior rather than a hard fixed-rank constraint.

5 Experimental Setup

Task. We use modular addition:

$$c = (a + b) \bmod p, \quad (11)$$

with $p = 113$ unless otherwise stated. The full dataset consists of all ordered pairs (a, b) .

Model. The model is a 2-layer Transformer with model dimension $d_{\text{model}} = 128$ and 4 attention heads. Pre-RMSNorm and QK-Norm are enabled to emphasize scale-invariant attention components.

Placeholder for phase diagram: L2 decay vs. Low-Rank Decay across training data fraction and decay coefficient.

Figure 1: Preliminary phase diagram of modular-addition generalization. Replace this placeholder with the exported figure from the experiment notebook.

Placeholder for rank-collapse dynamics: stable rank and test accuracy over training steps.

Figure 2: Preliminary rank-collapse dynamics. Replace this placeholder with stable-rank traces for L2 and LRD runs.

Baselines. We compare standard Frobenius weight decay (L2/AdamW-style decay) against decoupled Low-Rank Decay. In the current preliminary experiments, LRD is primarily applied to Query and Key matrices, where QK-Norm makes the scale-invariance assumption most direct. [TODO: Specify exact parameter groups used in the final code.]

Metrics. We track train accuracy, test accuracy, loss, weight norms, gradient norms, singular values, and stable rank. For a matrix W , stable rank is defined as

$$\text{srank}(W) = \frac{\|W\|_F^2}{\|W\|_2^2}. \quad (12)$$

We report stable rank for Query/Key matrices during training.

Grid. We sweep training data fraction and regularization strength to construct phase diagrams of generalization. A run is counted as successful grokking if it reaches test accuracy above 99%. [TODO: Add number of seeds and confidence intervals.]

6 Preliminary Results

6.1 Phase diagram of generalization

Figure 1 is a placeholder for the phase diagram. Preliminary results suggest that standard L2 decay exhibits a generalization boundary around roughly half of the full data, while LRD expands the successful grokking region to lower data fractions. In the current draft, we treat these findings as preliminary because multi-seed variance, learning-rate matching, and stronger baselines are still needed.

6.2 Rank collapse dynamics

Figure 2 is a placeholder for effective-rank traces. In early experiments, LRD causes the stable rank of Query/Key matrices to decrease sharply before the rise in validation accuracy. This temporal ordering is consistent with the hypothesis that spectral compression precedes algorithm discovery. However, temporal precedence alone does not prove causality; causal interventions are needed.

7 Discussion

7.1 LRD as soft low-rank selection

Compared with hard low-rank parameterizations such as LoRA, LRD does not fix the rank in advance. Instead, it applies continuous pressure on the singular spectrum. If a singular direction is useful for the task, the task gradient can counteract the nuclear-norm shrinkage. If a direction primarily supports memorization, it is more likely to be removed by the decay. This is the intended mechanism, but it remains to be tested with causal interventions.

7.2 Connection to geometry-preserving model merging

A natural follow-up is to compare spectral-shaping during training with geometry preservation during model merging. Methods such as orthogonal merging preserve geometric structure after task-specific capabilities have formed. LRD instead reshapes the spectrum while the model is discovering a circuit. This suggests a possible phase distinction: during training, it may be useful to collapse memorization-heavy spectral directions; after grokking, it may be useful to preserve the resulting low-dimensional algorithmic structure during merging.

One concrete future experiment is to train multiple small models that grok related modular tasks and then merge them using weight averaging versus an OrthoMerge-style orthogonal/residual decomposition. If grokked circuits are more geometrically composable than memorization solutions, LRD-trained models may merge more robustly.

7.3 Limitations

This draft has several important limitations:

1. Current evidence is based mainly on modular addition and should be extended to modular multiplication, permutation composition, and group operations.
2. Multi-seed results and error bars are needed to rule out instability or cherry-picking.
3. Learning-rate and effective-learning-rate controls are needed, since weight decay can indirectly change optimization speed in scale-invariant networks.
4. Exact SVD-based nuclear decay should be compared against Newton–Schulz LRD on small models.
5. Causal tests are needed to establish whether rank collapse causes grokking rather than merely correlating with it.

8 Conclusion

We presented Low-Rank Decay, a nuclear-norm-like spectral regularizer for studying grokking in scale-invariant Transformers. The central idea is that, when normalized layers remove sensitivity to radial weight scale, useful regularization should act on the singular spectrum rather than only on Frobenius magnitude. Preliminary modular-addition experiments suggest that LRD induces rank collapse before delayed generalization and may expand the data-scarce grokking regime. The current evidence is promising but incomplete; the next step is to establish robustness, perform causal interventions, and test whether grokked low-rank circuits can be preserved and composed through geometry-aware model merging.

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